

NgTrDat@cv-terminal:~

NgTrDat

Wed 8 Apr - 17:05

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fastfetch -c \$HOME/.config/fastfetch/CV.jsonc



NgTrDat@cv-terminal

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Name       : Nguyễn Trọng Đạt
Age-Gender : 22 - Male
Role       : Computer Engineering
Study      : Ho Chi Minh University of Technology
Focus      : Robotics • AI • Drones • Embedded Systems
Location   : Ho Chi Minh City, Vietnam
Email      : trongdat4work@gmail.com
Languages  : C/C++, Python, Shell script
Tools      : VSCode, Git, ROS2, Gazebo
English    : TOEIC L&R 970/990
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NgTrDat

Wed 8 Apr - 17:06

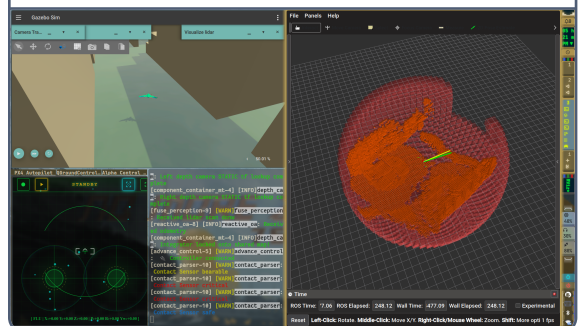
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./script/ShowMyCV.sh

Professional_objective.txt

- Computer Engineering student focused on robotics, embedded systems, and autonomous drone software. Experienced with ROS2, PX4, Gazebo, and C++ development for real-time systems and simulation environments.
- Experienced in collaborative development environments using Git; proficient at reading, understanding, and contributing safely to shared codebases.
- Seeking a part-time engineering role, potentially transitioning to full-time after graduation.

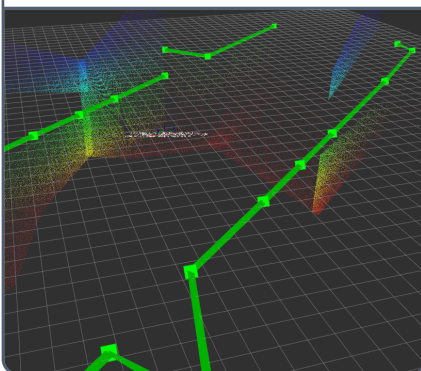
Drone_Simulation_Preview.png



Education.log

[Study] Ho Chi Minh University of Technology (BKU)
[Major] Bachelor of Engineering in Computer Engineering (Expected graduation 2027)
[GPA] 2.9/4.0
[Core Coursework] • Embedded System • Operating System • Machine Learning
• Microprocessors-Microcontrollers
• IoT-Oriented Logic Design Project
• AI-Oriented Multidisciplinary Project
[Passion Project] Developed a ROS2 and PX4 drone simulation pipeline in Gazebo to train AI models for VTOL acrobatic flight.

LIDAR_depth_scan.png



Skills.md

```
1 Programming:
2   • Most proficient with C++, Python and Rust
3   • Familiar with Java, Go, Zig
4 Robotics & Middleware ecosystem:
5   • ROS2 (Jazzy) • Gazebo Harmonic
6   • PX4 Autopilot • MicroXRCE-DDS
7 AI, Machine Learning and Data Engineering:
8   • Latent Diffusion Learning • Robotic data processing
9   • Reinforcement Learning • AI Edge Computing
10 Development tools: • VScode • Git • Linux OS
```

Contacts.txt

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✉ Email : trongdat4work@gmail.com
🌐 Github: github.com/LemonKronos
🔗 LinkIn: www.linkedin.com/in/trong-dat
* All source code below is available on GitHub

About_me.txt

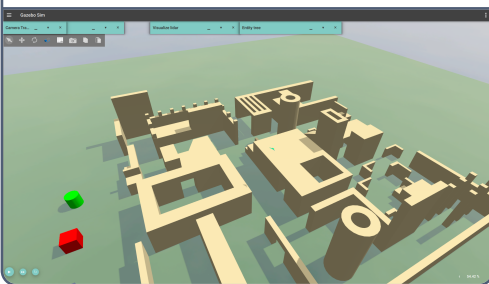
- I am a Drone enthusiast
- My goal is to become a Robotics Engineer
- Open-source mindset, member of Linux community
- Strong technical communicator, ready to collaborate in international teams

Multidisciplinary_Project_AI_Oriented.md

Project: Full-Pipeline Acrobatic Drone based on Latent Imitation Learning

- **Latent Imitation Learning:** Trains an AI model to imitate expert pilot behavior using latent space representations and performance-based evaluation, enabling policies that can potentially surpass the original demonstrations.
- **Goal:** framework designed to act as an "AI Co-pilot," denoising erratic, suboptimal human inputs into safe, expert-level 3D trajectories.
- **Main components:** Gazebo Harmonic for simulation and physics, PX4 Autopilot for high-frequency flight control (400Hz), Micro XRCE-DDS for PX4-ROS2 communication, and ROS2 as the offboard middleware handling sensors, messaging, and AI integration (80Hz). The acrobatic AI runs as a Python ROS2 node at a lower control frequency (0.5Hz).
- **Principle:** Inputs include raw human control commands, telemetry data, and environmental perception data (vision and 3D sensing). The AI generates an intended trajectory through the environment, which is validated by a safety and obstacle-avoidance layer before being executed by PX4 and translated into motor commands.

Test_Control_World.png



VTOL.png

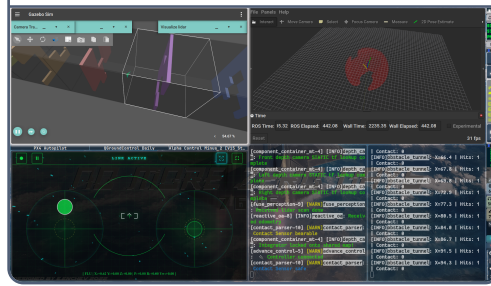


Internship_Work.md

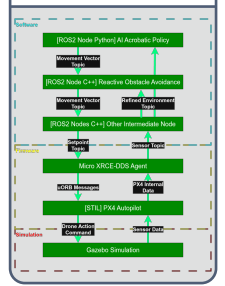
Project: Cross-Platform ITAM Endpoint Device Agent

- **Goal:** Develop a lightweight, memory-safe IT asset monitoring daemon for Linux and Windows.
- **Main components:** Rust, Tokio, Unix Domain Sockets, Min-Heap scheduler, flamegraph, hyperfine.
- **Principle:** Architecture is decoupled into discrete binaries (launcher, worker, fetcher) linked by IPC. This isolates faults—a fetcher panic will not crash the main daemon. Precise task scheduling is managed via a custom Min-Heap scheduler.
- **Execution:** Built the multi-binary Rust agent handling OS-specific system calls. Implemented the IPC layer for data passing and profiled memory/CPU bottlenecks using flamegraphs and hyperfine.
- **Results:** Successfully delivered a functioning cross-platform agent capable of scheduled data fetching and resilient inter-process communication without memory leaks.
- **Summarize:** Delivered a foundational MVP, successfully demonstrating Rust's utility as a highly effective choice for building memory-safe, low-overhead system architectures.

Test_non-AI_behavior.png



Pipeline.png



Multidisciplinary_Project_AI_Oriented.md

- **Execution:** Collected expert control inputs and telemetry data in Gazebo to build a training dataset. The simulation environment consisted of a procedurally generated obstacle-tunnel world. Models were trained offline and evaluated through both offline testing and real-time user-controlled flight scenarios.
- **Results:** Successfully validated the non-AI control pipeline, including low-level obstacle avoidance, achieving autonomous flights of up to 1 km without collisions. However, the AI-controlled system produced noisy outputs and occasionally conflicted with user inputs, resulting in unstable flight behavior. Simulated latency average 290us
- **Diagnosed:** Early model outputs revealed limitations in the design and data flow between high and low layers, indicated the need for an improved model and larger training datasets.
- **Future:** Currently redesigning the pipeline with a focus on low-level data processing, system integration, and control structures to support more dynamic acrobatic maneuvers.
- **Summarize:** Demonstrated the feasibility of the overall control pipeline. Further refinement of both the AI and robotics architectures is needed.

Capstone_Project_Plan.md

Project: VTOL Acrobatic Control & Twin Obstacle Avoidance Architecture

- Goals:** Design an AI-driven control pipeline that explores and exploits the highly non-linear flight envelope of VTOL while maintaining deterministic spatial safety.
- Main components:** C++, ROS2 (Jazzy), PyTorch (RL), Mixture of Experts (MoE), Voxblox (ESDF), Gazebo Harmonic, PX4 Autopilot.
- Principle:** Engineered a "Twin OA" architecture. The Acrobatic OA utilizes a Reinforcement Learning Mixture of Experts (MoE) network—featuring hardware-specific experts and a mixing layers—to generate complex actuator sequences and predicted states. These probabilistic AI outputs are strictly validated by the Reactive OA, which performs deterministic collision checks using a continuous Voxblox ESDF map. Validated states bypass the rigid PX4 mixer for direct acrobatic execution, while rejected states instantly trigger a spatial safety fallback.
- Summarize:** Proposing a novel, mathematically validated AI architecture designed to safely push VTOL hardware to its aerodynamic limits without relying on static control mixers.